

Supporting Information

Machine Learning-Driven Prediction of Dose-Linear Pharmacokinetics: Utilizing Molecular Descriptors to Guide Formulation Strategy

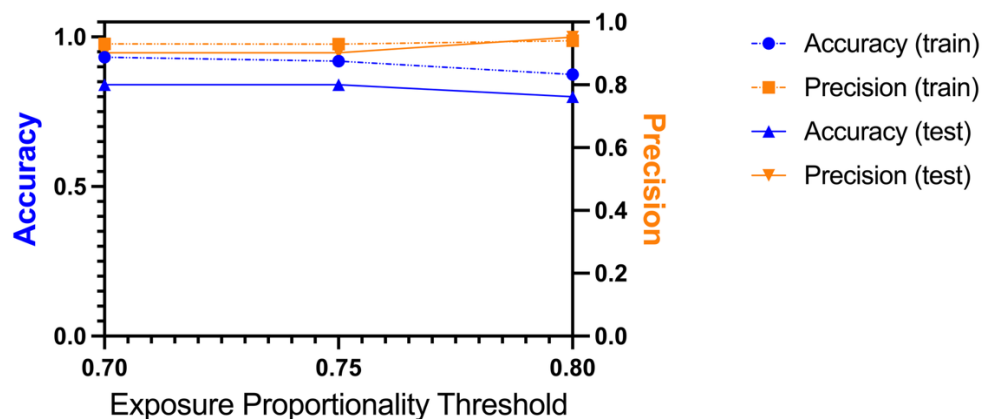
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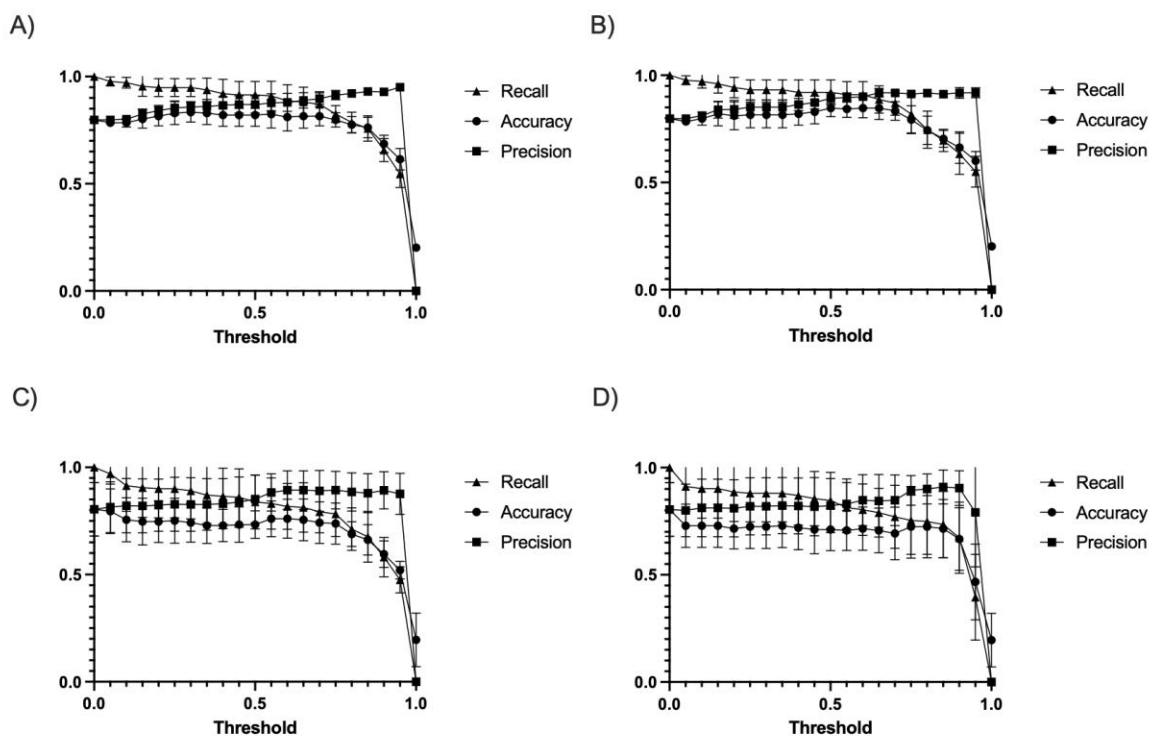
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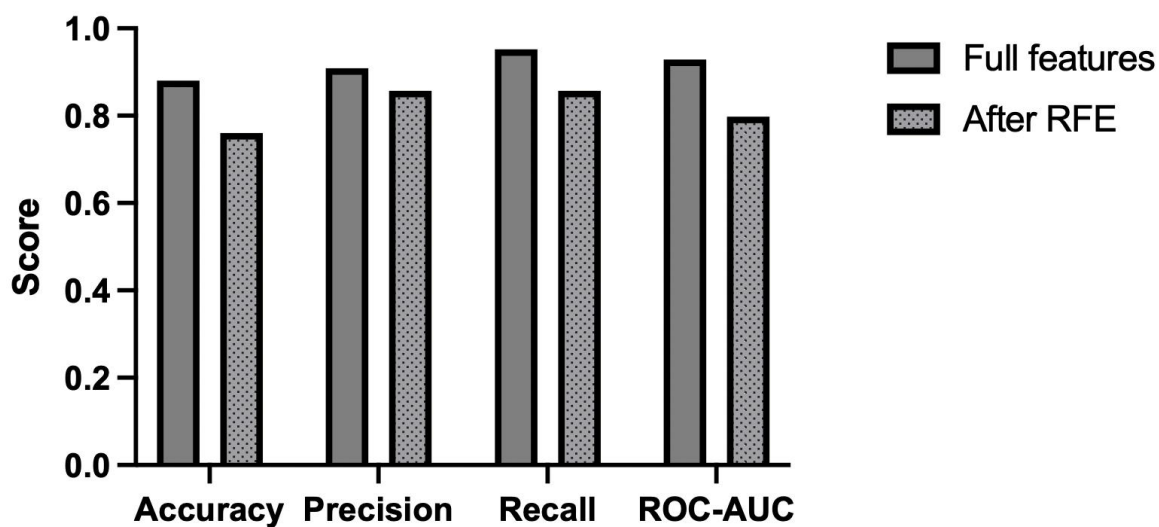
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Supplementary Figure 1: Model performance of the training and test datasets measured by Accuracy (left y-axis) and Precision (right y-axis) across three potential exposure proportionality thresholds (0.70, 0.75, and 0.80)



Supplementary Figure 2: ROC curves for LASSO model for (A) Random Split full descriptors (B) Random Split RDKit descriptors (C) Scaffold Split full descriptors (D) Scaffold Split RDKit descriptors



Supplementary Figure 3: Comparison model performance metrics (Accuracy, Precision, Recall, and ROC-AUC) calculated for the LASSO model with the full feature set and the optimized features after recursive feature elimination (RFE)

Supplementary Table 1: Physicochemical property distribution of the dataset

descriptor	min	q1	median	q3	max
MolWt	275.74	391.73	457.11	508.52	736.12
MolLogP	0.76	2.48	3.46	4.14	7.27
TPSA	36.10	75.46	95.86	110.52	141.43
NumHDonors	0	1	2	2	4
NumHAcceptors	2	5.75	7	8	12
NumRotatableBonds	2	4	5	6	12
NumAromaticRings	1	2	3	3.25	6
MolMR	72.10	102.77	120.41	133.25	175.24
HeavyAtomCount	19	28	32	35	51
RingCount	2	4	5	5	8
NumHeteroatoms	3	8	10	11	18
FractionCSP3	0.04	0.26	0.42	0.50	0.62
Properties cLogP	0.89	2.27	2.80	3.62	6.50
Properties Charge	0	0	0	1	2
Properties Rings	2	3	3	4	7
Properties AromaticRings	1	2	3	3.25	6
Properties H_polar	0	2	2	3	7

Properties N+O	3	6	8	9	13
Properties NH+OH	0	1	2	3	5
TPSA_x	36	74.75	95	110.25	141
TPSA_y	36.10	75.46	95.86	110.52	141.43
RotBonds	2	4	5	6	12
Predicted LogSolubility	0.91	6.94	25.78	67.17	152.92
Predicted LogSolubility FaSSIF	12.10	41.11	75.06	112.43	156.72
Solubility [ug/mL] FaSSIF	7	15.25	48.75	76	102
PBS Solubility [ug/mL]	0.45	2	18.50	42.25	104
Dose Number_FaSSIF	51.55	454.55	1100.92	1756.73	20000
Dose Number_PBS	48.08	811.00	2683.78	21279.76	200000
Hepatocyte Stability T1/2 [min] Mouse	6	137	401.75	540	18909
Liver Microsome Stability CLint [ml/min/kg] Mouse	17	53	89	205	2370
Liver Microsome Stability T1/2 [min] Mouse	2.65	34	73	165	969
Permeability Papp (A to B) [10E-06 cm/s] gMDCKI	0.26	1.65	3.80	9.65	26
Permeability Papp (A to B) [10E-06 cm/s] gMDCKI-MDR1	0.01	0.07	0.39	2.45	21
Permeability Papp Ratio [10E- 06 cm/s] gMDCKI-MDR1	0.59	7.88	42	206.75	1982
Hepatocyte Stability CLhep [ml/min/kg] Mouse	4.30	21	25.25	49	87
BalabanJ	1.08	1.36	1.57	1.77	2.27
BertzCT	615.04	974.38	1131.88	1319.65	2104.85
Chi0	13.46	19.83	22.62	24.84	36.25
Chi0n	10.24	15.51	18.10	20.17	27.42
Chi0v	10.99	16.05	18.57	21.05	28.18
Chi1	9.08	13.37	15.27	16.99	24.06
Chi1n	5.90	8.75	10.70	12.33	16.34
Chi1v	6.28	9.53	11.33	13.03	16.71
Chi2n	4.58	6.80	8.33	10.19	13.93
Chi2v	4.99	7.80	8.99	10.92	14.31
Chi3n	3.18	4.49	5.91	7.62	10.82
Chi3v	3.59	5.33	6.48	8.06	11.18
Chi4n	1.90	3.06	4.21	5.69	8.31
Chi4v	2.35	3.55	4.69	6.05	8.63

EState_VSA1	0	5.60	11.54	23.71	66.37
EState_VSA10	0	5.11	11.61	17.97	32.35
EState_VSA11	0	0	0	0	4.39
EState_VSA2	0	11.15	17.64	29.24	79.31
EState_VSA3	0	12.92	24.60	37.65	74.19
EState_VSA4	0	12.43	27.46	43.64	65.22
EState_VSA5	0	0	11.13	18.66	74.12
EState_VSA6	0	6.07	12.26	22.95	68.50
EState_VSA7	0	4.90	12.99	28.25	64.34
EState_VSA8	4.98	17.31	21.14	30.06	69.02
EState_VSA9	0	4.74	11.53	15.95	42.26
FpDensityMorgan1	0.88	1.09	1.18	1.27	1.40
FpDensityMorgan2	1.52	1.86	1.94	2.04	2.24
FpDensityMorgan3	2.14	2.52	2.63	2.76	3
HallKierAlpha	-5.12	-3.38	-2.91	-2.43	-1.58
HeavyAtomMolWt	261.62	373.30	432.33	485.46	703.87
Ipc	3.11 E+4	2.24 E+6	1.88 E+7	1.36 E+7	1.57 E+11
Kappa1	12.05	18.74	21.60	23.86	33.75
Kappa2	4.28	7.35	8.32	9.41	13.67
Kappa3	1.86	3.44	4.15	4.81	7.03
LabuteASA	114.08	162.33	186.72	210.54	291.86
MaxAbsEStateIndex	4.85	12.19	13.22	14.37	17.06
MaxAbsPartialCharge	0.31	0.38	0.39	0.48	0.57
MaxEStateIndex	4.85	12.19	13.22	14.37	17.06
MaxPartialCharge	0.13	0.23	0.26	0.39	0.57
MinAbsEStateIndex	0.00	0.03	0.07	0.16	0.53
MinAbsPartialCharge	0.13	0.23	0.26	0.34	0.49
MinEStateIndex	-4.96	-4.10	-0.82	-0.45	0.42
MinPartialCharge	-0.50	-0.46	-0.39	-0.38	-0.28
NHOHCount	0	1	2	3	5
NOCOUNT	3	6	8	9	13
NumAliphaticCarbocycles	0	0	0	1	4
NumAliphaticHeterocycles	0	0	1	2	3
NumAliphaticRings	0	1	2	3	6
NumAromaticCarbocycles	0	1	1	1	3
NumAromaticHeterocycles	0	1	2	2	4
NumRadicalElectrons	0	0	0	0	0
NumSaturatedCarbocycles	0	0	0	1	4
NumSaturatedHeterocycles	0	0	1	2	3
NumSaturatedRings	0	0	1	2	5
NumValenceElectrons	98	146	170	188	270
PEOE_VSA1	0	10.46	15.29	20.23	30.58

PEOE_VSA10	0	6.09	12.08	19.31	47.05
PEOE_VSA11	0	0	5.14	10.83	23.07
PEOE_VSA12	0	0	5.91	5.95	17.72
PEOE_VSA13	0	0	0	5.91	21.36
PEOE_VSA14	0	0	0	6.11	12.27
PEOE_VSA2	0	4.79	4.94	9.81	24.68
PEOE_VSA3	0	8.78	13.40	15.59	37.13
PEOE_VSA4	0	0	4.64	10.20	23.14
PEOE_VSA5	0	0	0	1.27	24.51
PEOE_VSA6	0	5.84	12.13	23.73	60.93
PEOE_VSA7	6.07	31.19	42.42	51.16	85.90
PEOE_VSA8	0	24.46	35.21	47.90	68.79
PEOE_VSA9	0	12.11	22.19	31.15	54.22
SMR_VSA1	0	9.57	14.51	22.66	41.48
SMR_VSA10	0	17.65	27.56	35.14	66.28
SMR_VSA2	0	0	0	0	5.26
SMR_VSA3	0	14.78	20.01	24.40	38.87
SMR_VSA4	0	0	0	5.73	12.97
SMR_VSA5	0	18.99	32.95	57.22	75.92
SMR_VSA6	0	7.09	18.91	37.11	89.59
SMR_VSA7	16.08	36.33	51.91	58.90	105.77
SMR_VSA8	0	0	0	0	0
SMR_VSA9	0	0	11.13	11.96	39.78
SlogP_VSA1	0	10.05	10.47	15.20	25.84
SlogP_VSA10	0	5.82	11.77	22.28	43.80
SlogP_VSA11	0	0	0	5.75	23.00
SlogP_VSA12	0	0	11.34	11.60	27.53
SlogP_VSA2	15.88	34.21	47.72	63.64	130.61
SlogP_VSA3	0	6.42	10.97	16.21	36.47
SlogP_VSA4	0	0	5.82	11.41	25.48
SlogP_VSA5	4.88	18.64	32.62	55.68	83.38
SlogP_VSA6	5.38	24.40	30.59	42.99	71.85
SlogP_VSA7	0	0	0	5.02	10.18
SlogP_VSA8	0	0	11.15	22.03	38.88
SlogP_VSA9	0	0	0	0	0
VSA_EState1	0	6.77	16.20	42.77	101.80
VSA_EState10	0	0	1.12	6.12	12.55
VSA_EState2	3.98	16.82	21.62	27.15	58.03
VSA_EState3	-0.51	3.07	6.56	13.74	27.47
VSA_EState4	-1.84	1.32	2.92	6.21	12.90
VSA_EState5	-4.64	-0.64	0	1.13	2.96
VSA_EState6	-1.51	4.01	6.88	8.33	24.24

VSA_EState7	-4.80	0.30	3.05	4.52	9.31
VSA_EState8	-1.00	1.87	3.84	6.82	10.94
VSA_EState9	-4.41	0	0	0	6.58
fr_Al_COO	0	0	0	0	0
fr_Al_OH	0	0	0	1	2
fr_Al_OH_noTert	0	0	0	0	2
fr_ArN	0	0	0	0	2
fr_Ar_COO	0	0	0	0	1
fr_Ar_N	0	2	2	4	7
fr_Ar_NH	0	0	0	0	1
fr_Ar_OH	0	0	0	0	1
fr_COO	0	0	0	0	1
fr_COO2	0	0	0	0	1
fr_C_O	0	0	1	1	4
fr_C_O_noCOO	0	0	1	1	4
fr_C_S	0	0	0	0	0
fr_HOCCN	0	0	0	0	1
fr_Imine	0	0	0	0	1
fr_NH0	0	2	4	5	10
fr_NH1	0	0.75	1	2	3
fr_NH2	0	0	0	0.25	2
fr_N_O	0	0	0	0	1
fr_Ndealkylation1	0	0	0	0	1
fr_Ndealkylation2	0	0	0	0	1
fr_Nhpyrrole	0	0	0	0	1
fr_SH	0	0	0	0	0
fr_aldehyde	0	0	0	0	0
fr_alkyl_carbamate	0	0	0	0	1
fr_alkyl_halide	0	0	0	2.25	5
fr_allylic_oxid	0	0	0	0	2
fr_amide	0	0	1	1	4
fr_amidine	0	0	0	0	0
fr_aniline	0	1	1	2	6
fr_aryl_methyl	0	0	0	1	2
fr_azide	0	0	0	0	0
fr_azo	0	0	0	0	0
fr_barbitur	0	0	0	0	0
fr_benzene	0	1	1	1	3
fr_benzodiazepine	0	0	0	0	0
fr_bicyclic	0	1	1	2	5
fr_diazo	0	0	0	0	0
fr_dihydropyridine	0	0	0	0	0

fr_epoxide	0	0	0	0	0
fr_ester	0	0	0	0	0
fr_ether	0	0	1	1	4
fr_furan	0	0	0	0	0
fr_guanido	0	0	0	0	0
fr_halogen	0	0	1	3	7
fr_hdrzine	0	0	0	0	0
fr_hdrzone	0	0	0	0	0
fr_imidazole	0	0	0	0	2
fr_imide	0	0	0	0	1
fr_isocyan	0	0	0	0	0
fr_isothiocyan	0	0	0	0	0
fr_ketone	0	0	0	0	0
fr_ketone_Topliss	0	0	0	0	0
fr_lactam	0	0	0	0	0
fr_lactone	0	0	0	0	1
fr_methoxy	0	0	0	0	1
fr_morpholine	0	0	0	0	2
fr_nitrile	0	0	0	0	1
fr_nitro	0	0	0	0	0
fr_nitro_arom	0	0	0	0	0
fr_nitro_arom_nonortho	0	0	0	0	0
fr_nitroso	0	0	0	0	0
fr_oxazole	0	0	0	0	1
fr_oxime	0	0	0	0	0
fr_para_hydroxylation	0	0	0	0	2
fr_phenol	0	0	0	0	1
fr_phenol_noOrthoHbond	0	0	0	0	1
fr_phos_acid	0	0	0	0	0
fr_phos_ester	0	0	0	0	0
fr_piperdine	0	0	0	0	2
fr_piperzine	0	0	0	0	1
fr_priamide	0	0	0	0	1
fr_prisulfonamd	0	0	0	0	0
fr_pyridine	0	0	0	1	3
fr_quatN	0	0	0	0	0
fr_sulfide	0	0	0	0	0
fr_sulfonamd	0	0	0	0	1
fr_sulfone	0	0	0	0	1
fr_term_acetylene	0	0	0	0	0
fr_tetrazole	0	0	0	0	0
fr_thiazole	0	0	0	0	2

fr_thiocyan	0	0	0	0	0
fr_thiophene	0	0	0	0	1
fr_unbrch_alkane	0	0	0	0	1
fr_urea	0	0	0	0	1
qed	0.15	0.46	0.57	0.71	0.92
Liver Microsome Stability CLhep [ml/min/kg] Mouse	9.90	23	43.67	60	86.50
Hepatocyte Stability CLint [ml/min/kg] Mouse	27	27	45.25	119	2420
RO5	0	0	0	1	3